What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Histology of the Accessory Digestive Organs

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Adapted from: Anatomy and Histology of the Liver and Biliary Tree

Listen to this Brick

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Describe the histologic organization of the pancreas.
2. Describe the histologic organization of the liver, highlighting the different ways of examining liver microscopic structure.
3. Describe the tract and gallbladder.

CASE CONNECTION

A 13-year-old patient presents to the clinic for his annual check-up. Three years ago, he was diagnosed with type I diabetes mellitus after presenting with excessive thirst and urination. If a portion of his pancreas was biopsied, it would demonstrate a decrease in the number and size of the Islets of Langerhans with associated inflammation. What is the function of the Islets of Langerhans? Does that explain his disease?

Consider your answers to this question, and we will revisit the case at the end of the brick.

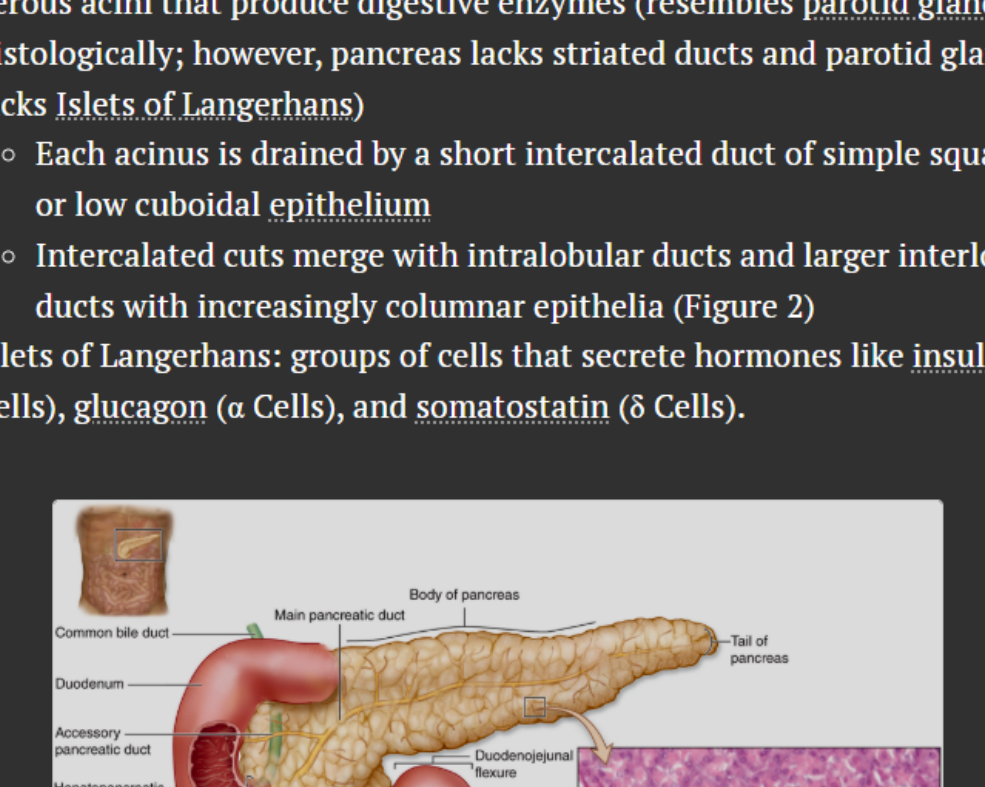
GO TO CONCLUSION

The organs associated with the digestive tract include the major salivary glands, pancreas, liver, and gallbladder. Products of these organs facilitate movement and digestion of food within the GI tract. The histology of the major salivary glands was previously covered in a brick on the oral cavity.

What is the Histologic Organization of the Pancreas (LO #1)?

The pancreas develops as a mixed exocrine-endocrine gland producing both digestive enzymes an logic structures (Figure 1):

- Serous acini that produce digestive enzymes (resembles parotid gland histologically; however, pancreas lacks striated ducts and parotid gland lacks Islets of Langerhans)
    - Each acinus is drained by a short intercalated duct of simple squamous or low cuboidal epithelium
    - Intercalated cuts merge with intralobular ducts and larger interlobular ducts with increasingly columnar epithelia (Figure 2)
- Islets of Langerhans: groups of cells that secrete hormones like insulin (β Cells), glucagon (α Cells), and somatostatin (δ Cells).



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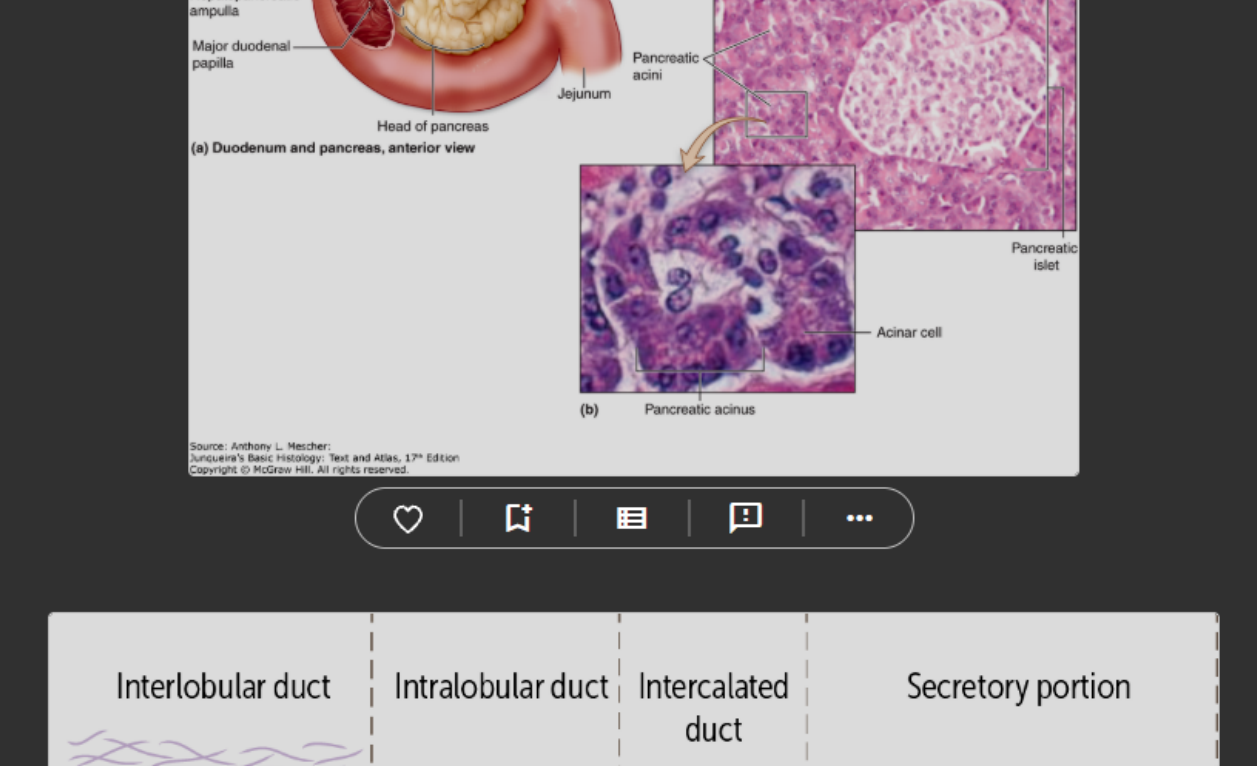


Figure 2. Ductal Organization of the Pancreas

What are the two main histologic structures of the pancreas?

What Is the Histologic Organization of the Liver (LO #2)?

The liver is the largest internal organ, with numerous functions including production of bile (substance required for breakdown of fats in the duodenum), nutrient absorption, detoxification of substances, storage or glucose, and production of major plasma proteins (this is not a complete list).

The key cells of this organ are **hepatocytes**. The liver's unique histologic organization and microvasculature allows hepatocytes to perform their diverse functions. Hepatocytes are large cuboidal or polyhedral epithelial cells, with big round central nuclei and eosinophilic cytoplasm rich in mitochondria (Figure 3).

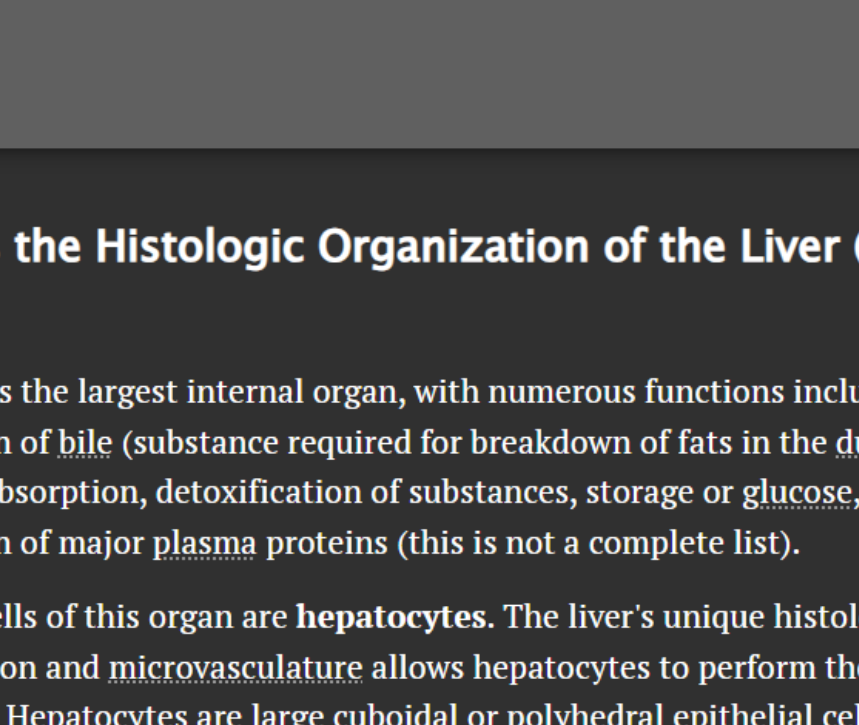


Figure 3. Hepatocytes (H) form branching, irregular plates separated by venous sinusoids (S).

Hepatic Lobules

The liver is organized in small building blocks of liver tissue called hepatic lobules. These lobules contain the hepatocytes and are the functional units of the liver:

- The lobules are hexagonal.
- There are six portal triads at the periphery of each lobule, one triad at each corner.
- Each portal triad contains small branches of the portal vein, proper hepatic artery, and bile duct.
- In the center of each hepatic lobule is a central vein; later, they unite to form the hepatic vein that drains blood from the liver.

These structures are shown in Figure 4 and Figure 5.

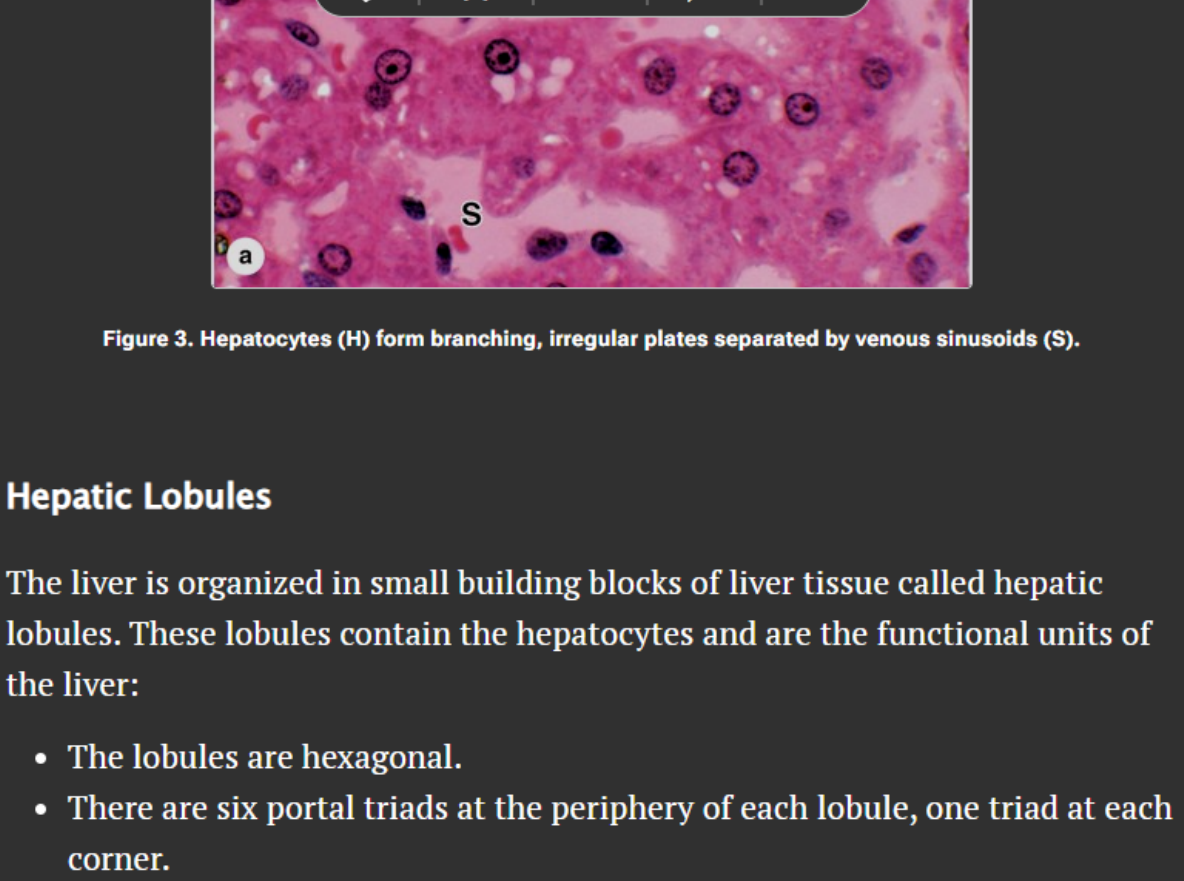


Figure 4

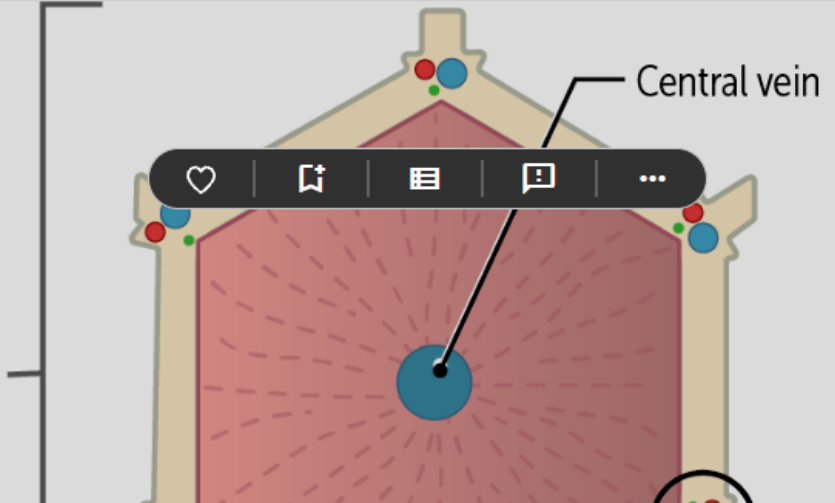
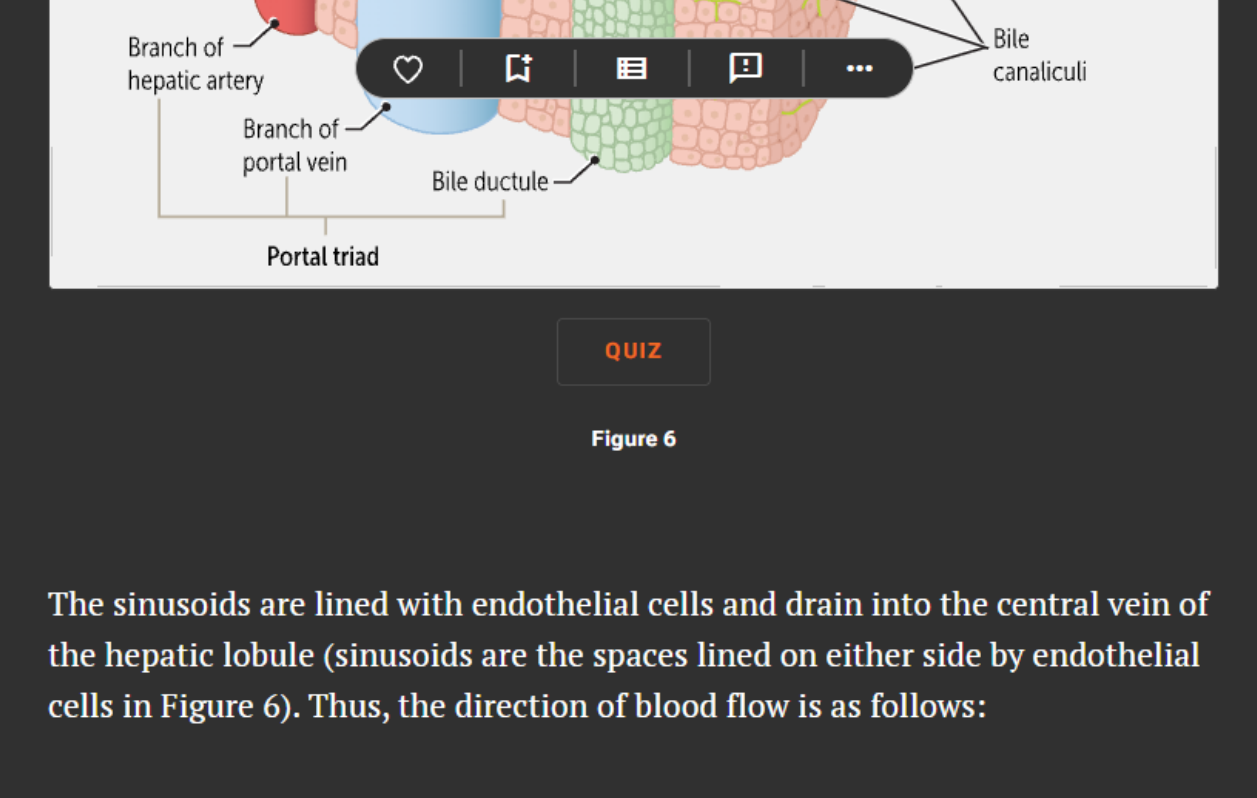


Figure 5. Hepatic lobule. Central vein (C) marks the center of the lobule, with one corner showing the portal triad: venule (V) branches of the portal vein, arteriole (A) branches of the hepatic artery, and small bile ductules (D).

Sinusoids

The portal triads are connected to each other by sinusoids. The sinusoids are blood vessels that contain numerous Kupffer cells—specialized macrophages that break down old red blood cells and have other immunologic functions (Figure 6).



QUIZ

Figure 6

The sinusoids are lined with endothelial cells and drain into the central vein of the hepatic lobule (sinusoids are the spaces lined on either side by endothelial cells in Figure 6). Thus, the direction of blood flow is as follows:

> Portal vein → Hepatocyte → Sinusoids → Central vein → Hepatic vein

The bile flow is:

> Hepatocytes → Bile canaliculi → Bile duct in the portal triad

Space of Disse

Between the sinusoid and the hepatocytes is the space of Disse, which contains stellate cells. These are normally quiescent, but during liver injury they proliferate and make collagen. They are therefore important in the scarring seen in chronic liver disease, called cirrhosis.

CLINICAL CORRELATION

Cirrhosis is the chronic liver scarring seen in patients with chronic alcohol overuse, chronic viral hepatitis, and other liver diseases. It leads to confusion, ascites (swelling of the peritoneal cavity), decreased hepatocyte function, and bleeding from congested veins lining the esophagus.

What is at the center of each hepatic lobule?

What Are the Functional Microscopic Zones of the Liver?

The hepatocytes look pretty much alike from the outside, but there is considerable specialization. The functions of hepatocytes nearer to the portal triad are different from those farther away. We describe this by identifying microscopic zones within the liver.

Understanding the zones starts with the hepatic acinus—a diamond-shaped clump of hepatocytes. Its corners are two portal triads and two central veins (Figure 7).

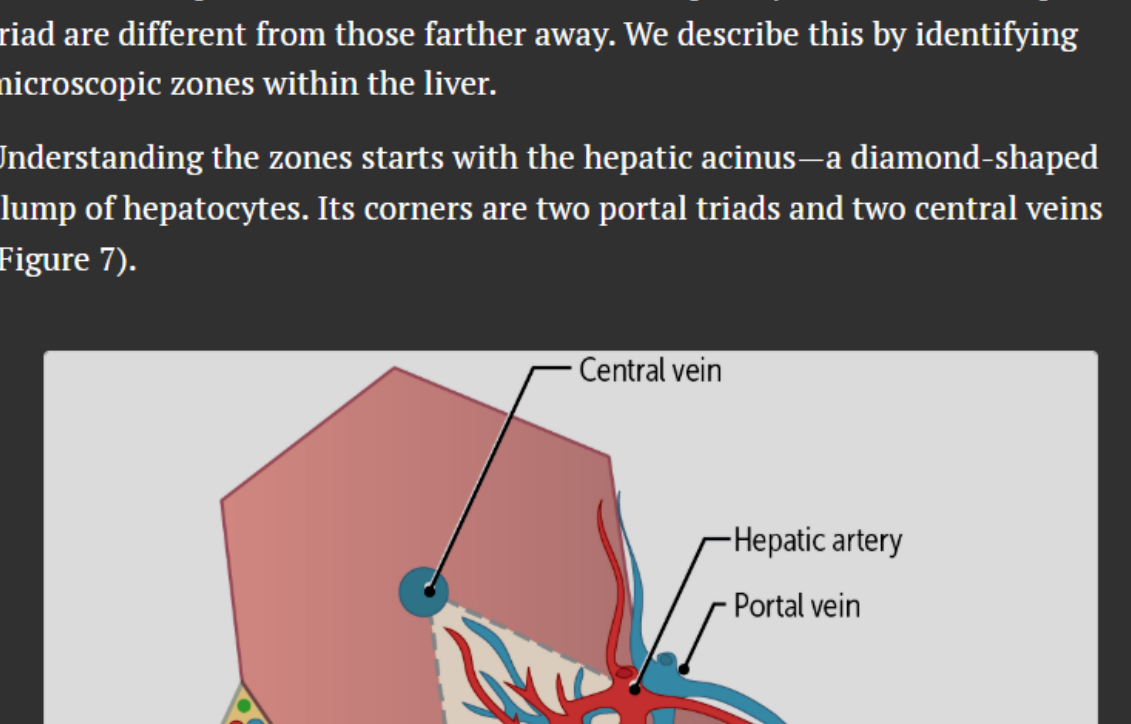


Figure 7

Each acinus is divided into three zones; the zone number reflects how close or far the hepatocytes within it are from the portal triad (Figure 8).

- Zone 1 (periportal zone) is the zone that's closest to the portal triad, where the arterial blood supply is located. This location makes it the most oxygenated zone, and zone 1 performs the oxygen-dependent processes of gluconeogenesis and cholesterol synthesis. As the first zone to interact with potentially harmful substances in the blood, this zone is the most susceptible to injury from some ingested toxins (i.e. cocaine, iron salts).
- Zone 2 (intermediate zone) is the middle zone and therefore is partially oxygenated. Its oxygen content is less than that of zone 1 but greater than zone 3.
- Zone 3 (pericentral or centrilobular zone) is the farthest from the portal triad. It contains the cytochrome P-450 system, the site of detoxification. This makes it susceptible to toxins formed by metabolic processing (i.e. acetaminophen, carbon tetrachloride). It is also the site of glycolysis and lipogenesis. Because it is the most poorly oxygenated zone, it is most susceptible to ischemic injury.

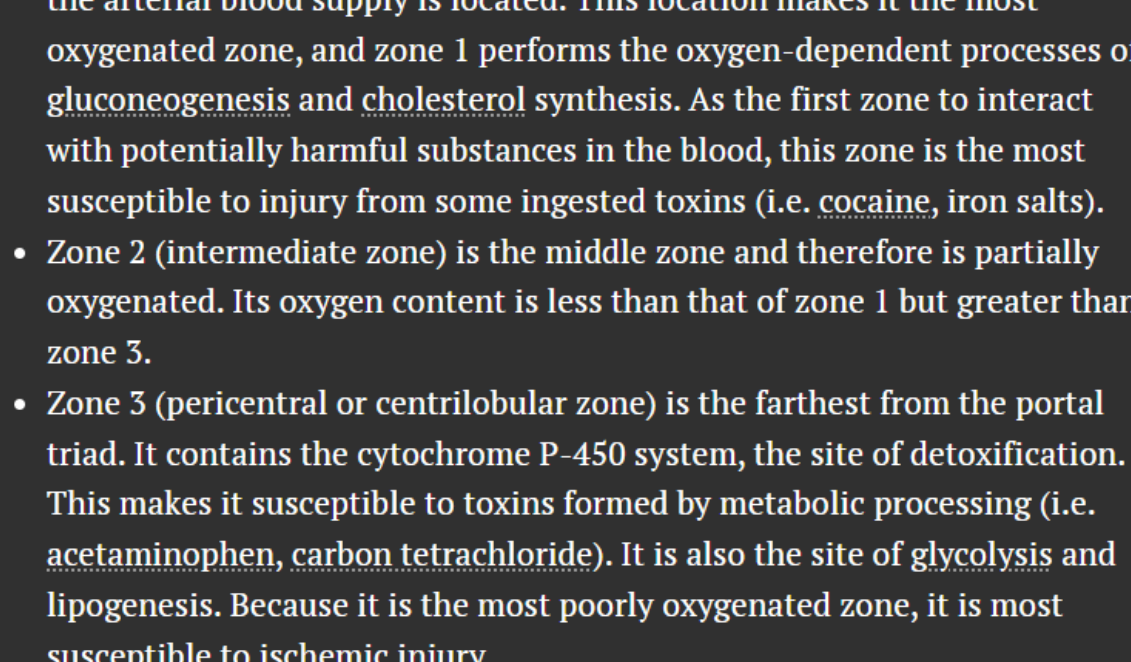


Figure 8

CLINICAL CORRELATION

Zone 1 is the earliest zone to be affected by viral hepatitis and ingested toxins, such as cocaine. Zone 2 is most affected by yellow fever. Zone 3 is the first zone to be affected by ischemia, most sensitive to metabolic toxins, and the site of hepatocyte damage from alcoholic hepatitis.

What is the Histologic Organization of the Gallbladder (LO #3)?

Bile produced by the hepatocytes flows through the bile canaliculi, bile ductules, and bile ducts. These bile ducts merge and ultimately form the common hepatic duc the gallbladder and continues to the duodenum as the common bile duct. Biliary structures after the bile canaliculi have an epithelial lining of simple cuboidal or columnar cells called cholangiocytes.

The gallbladder is a pear-shaped organ whose function is to store and concentrate bile.

Figure 9 outlines the basic structure of the gallbladder wall. Its wall consists of a highly folded mucosa lined by a simple columnar epithelium (arrows) overlying a lamina propria (LP), a muscularis (M) layer with bundles of muscle fibers oriented in all directions to facilitate emptying, and an external adventitia (A) where it sits against the liver (serosa where it's exposed).

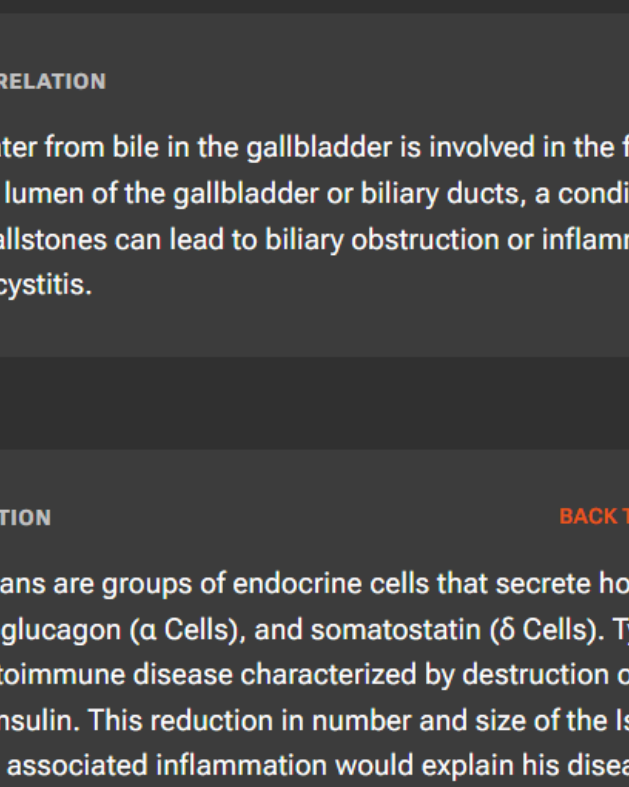


Figure 9

CLINICAL CORRELATION

Resorption of water from bile in the gallbladder is involved in the formation of gallstones in the lumen of the gallbladder or biliary ducts, a condition called cholelithiasis. Gallstones can lead to biliary obstruction or inflammation in acute or chronic cholecystitis.

CASE CONNECTION BACK TO INTRODUCTION

Islets of Langerhans are groups of endocrine cells that secrete hormones like insulin (β Cells), glucagon (α Cells), and somatostatin (δ Cells). Type I diabetes mellitus is an autoimmune disease characterized by destruction of pancreatic β cells that make insulin. This reduction in number and size of the Islets of Langerhans with associated inflammation would explain his disease.

Summary

- The pancreas is a mixed exocrine-endocrine gland with serous acini for digestive enzymes and Islets of Langerhans for hormone secretion.
- The liver is organized into hepatic lobules, with hepatocytes performing diverse functions ation.
- Hepatic lobules contain a central vein and portal triads, connected by sinusoids lined with Kupffer cells for immunologic functions.
- The liver has functional microscopic zones, with zone 1 being the most oxygenated and zone 3 being the site of detoxification.
- The gallbladder stores and concentrates bile, with a wall structure consisting of mucosa, muscularis, and adventitia or serosa.

Review Questions

1. **A 65-year-old male comes in to the clinic for a yearly physical, and the doctor finds that his liver is enlarged. The patient has suffered from alcohol use disorder for more than 15 years. He had abstained from alcohol use for 6 months but recently started drinking again. His tests show that his liver enzymes are very high. The doctor diagnoses him with alcoholic hepatitis. Which of the following liver areas would be most affected by his disease?**

A. ○ Common hepatic duct

B. ○ Intermediate zone (zone 2)

C. ○ Pericentral zone (zone 3)

D. ○ Periportal zone (zone 1)

E. ○ Portal triad

Explanation (requires co

2. **Which cells secrete insulin in the pancreas?**

A. ○ α cells

B. ○ β cells

C. ○ δ cells

D. ○ Acinar cells

E. ○ Hepatocytes

Explanation (requires correct answer)

3. **Which layer of the gallbladder wall facilitates emptying of bile?**

A. ○ Adventitia

B. ○ Lamina propria

C. ○ Mucosa

D. ○ Muscularis

Explanation (requires co

How would you rate this Brick?

References

- [redacted]

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ **Describe the histologic organization of the pancreas.**

02 ☐ **Describe the histologic organization of the liver, highlighting the different ways of examining liver microscopic structure.**

03 ☐ **Describ tract and gallbladder.**

References

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01 ☐ **Describe the histologic organization of the pancreas.**

02 ☐ **Describe the histologic organization of the liver, highlighting the different ways of examining liver microscopic structure.**

03 ☐ **Describe the histologic organization of the biliary tract and gallbladder.**

Objective Confidence: 0/3